PIPer: On-Device Environment Setup via Online Reinforcement Learning (& more)

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TL;DR: 🚀⚙️ **We use SFT and online RL with LLM reward to boost open-source LLMs' environment setup capabilities by +690%, pushing Qwen3-8B beyond 4× larger model at a fraction of the cost!**

Problem & Motivation

Environment Setup is important — but hard!

- Automating environment setup enables scaling execution-based SE benchmarks and assists developers with daily work
- LLMs struggle with environment setup (best success rate on EnvBench is 6.69% — achieved by GPT-4o in agentic scaffold)

Let's democratize environment setup with small local models!

- Reinforcement Learning with Verifiable Rewards (RLVR) shows promise in math and SE tasks
- RLVR needs reward functions that score model outputs and not supervised data

Challenges:

- No prior work applying RL to environment setup
- Execution-based rewards are computationally expensive as they require containerization

EnvBench Results

✨ **Key Results:**

- **Dramatic Improvements:** PIPer significantly outperforms base Qwen3-8B, approaching GPT-4o and Qwen3-32B performance with lower cost
- **No Memorization:** still good on validation set
- **Ablation:** PIPer > SFT-only or RL-only

🔎 Results

💸 **Efficiency:**

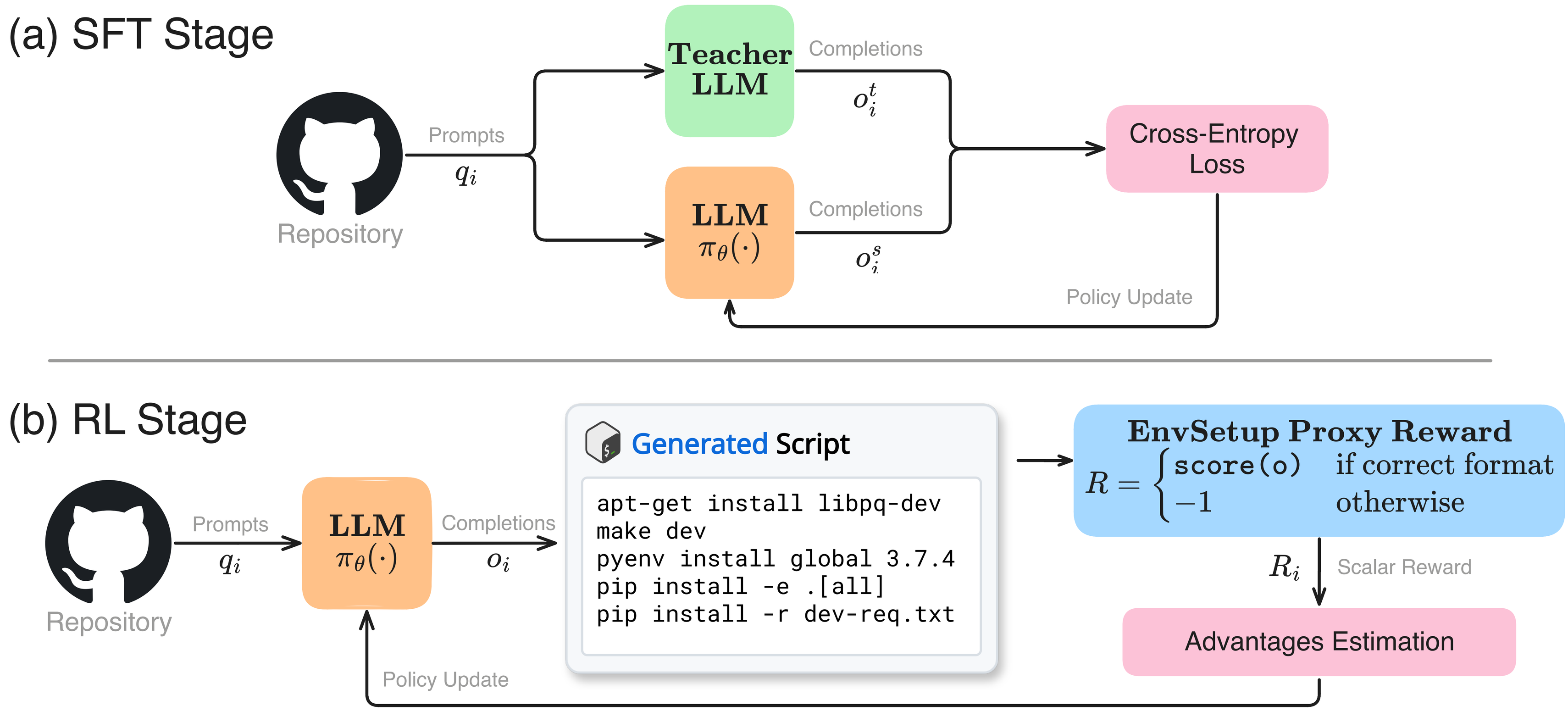
- Training: 4 hours on 1×H200 GPU for SFT; on 4×H200 GPUs for RL
- ~$250 API cost for LLM-as-a-Judge reward per RL training

🤫 **What didn't go into the paper:**

- GPT-4.1-mini and GPT-5-mini achieve VERY strong results on EnvBench, with pass@5 of 108 and 182 (still zero-shot!)

PIPer: SFT + RLVR with Proxy Reward

Base Model: Qwen3-8B (no thinking)
Scaffold: Zero-shot (one LLM request with repository context)
Data: 228 EnvBench-Python repositories (96 held-out as validation set)
Algorithm: SFT on 2500 trajectories with executable scripts from Qwen3-32B -> online RL with REINFORCE++
Reward: LLM-as-a-Judge where GPT-4.1 evaluates script correctness
Frameworks: LLaMA-Factory (SFT) & VeRL (RL)



Other Experiments

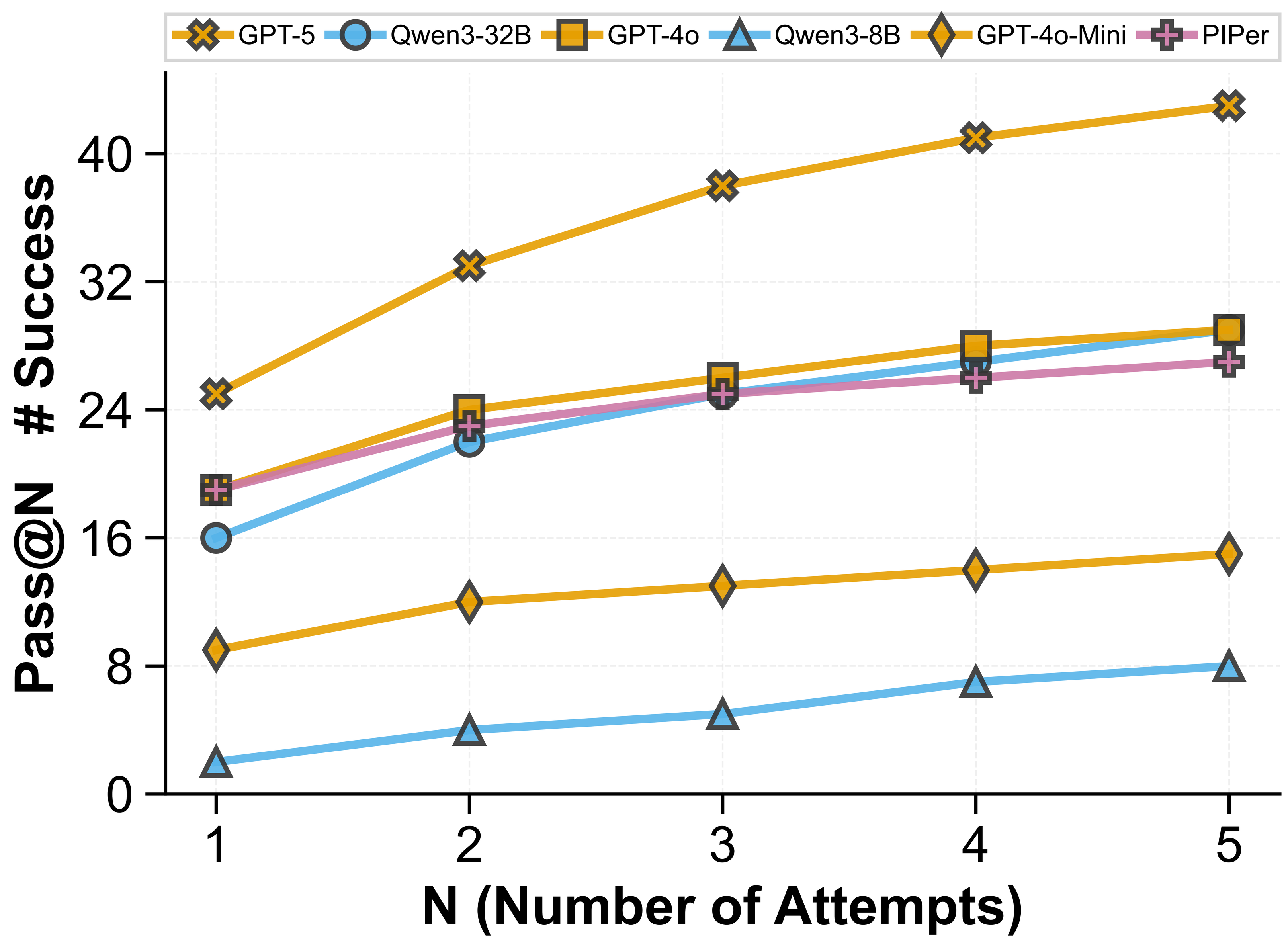
IDEGym: Containerized execution for RLVR training

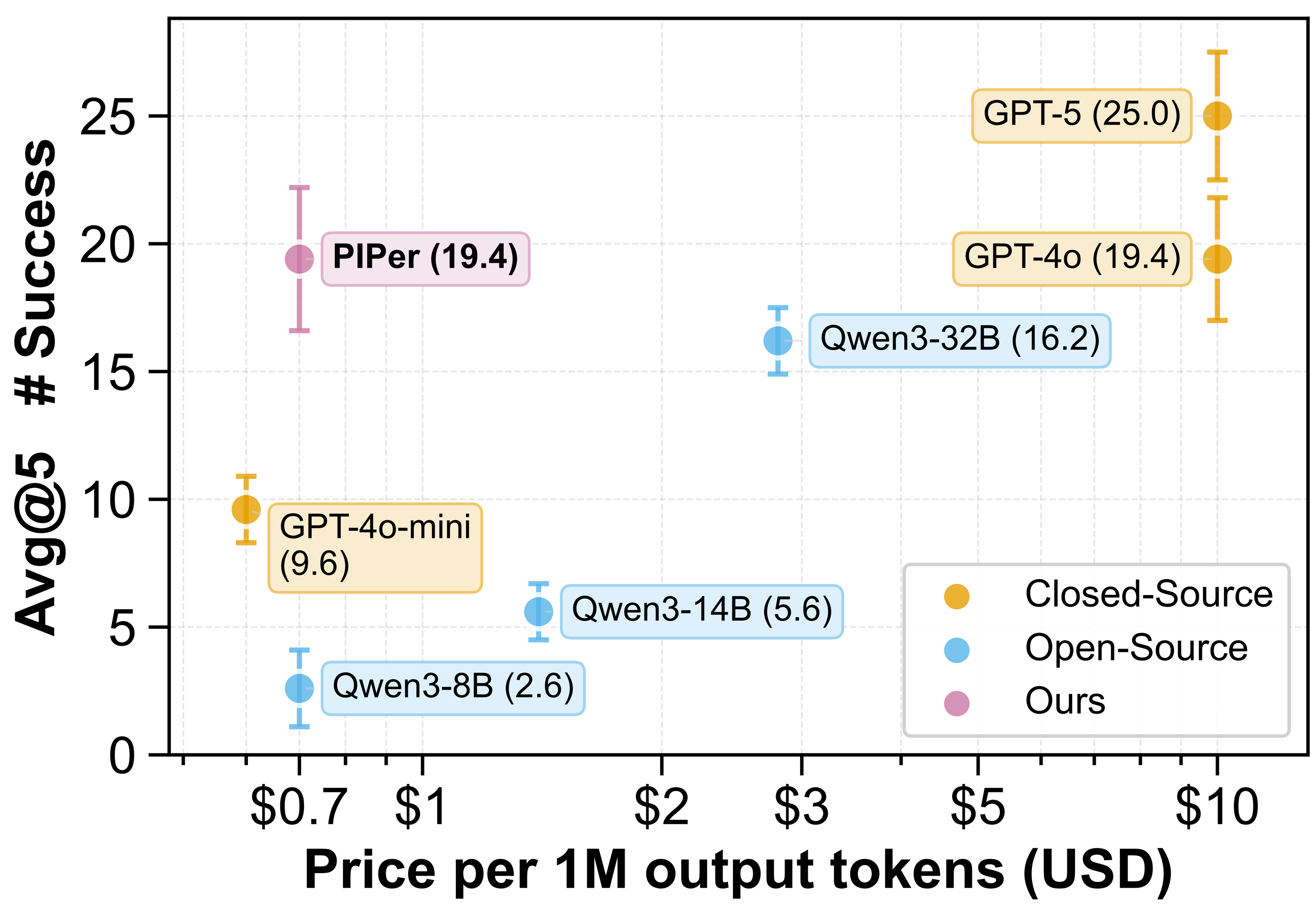
- ❌ Reward growth challenges remain

Proxy Rewards: LLM-Judge > Heuristics > ShellCheck

Multi-Turn: Underperformed zero-shot approach

❤️ IDEGym





Future Work

- Improvements for multi-turn agentic scaffolds
- Larger models, other model families, and reasoning LLMs
- Improvements for ground-truth execution reward
- Improvement of trajectory-aware rewards
- Move our approach to other software engineering tasks, such as SWE-bench-like problems.

Our Resources

Code | Models & Data | Preprint